Sahaj Bhadja

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Summary

Software Engineer with 2+ years of experience building **Python/FastAPI services**, **object-oriented applications**, automated tests, Linux workflows, secure financial systems, and maintainable production integrations.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Operating Systems, Software Engineering

Experience

USC Dornsife | *Software Engineer* | *Los Angeles, CA* **September 2025 – Present**

- Tripled concurrent throughput across 1K+ AI jobs by restructuring **FastAPI background services** with explicit state transitions, failure isolation, recoverable processing, and observable workflows under sustained production load reliably.
- Achieved 99%+ reliability across 5K+ monthly jobs through **automated validation and reconciliation**, error handling, cloud storage, CI/CD releases, and stable production observability for durable downstream workflows reliably.

Grad8 | *Software Engineer Intern* | *Los Angeles, CA* **June 2025 – August 2025**

- Reduced feedback turnaround by 60% through a **production Python platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, AWS RDS, secure ingestion, and CI/CD workflows.
- Processed 1K+ files per batch using **Python multithreading**, validating heterogeneous inputs while isolating failures and preserving responsive customer workflows during concurrent model execution and production traffic reliably.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) | *India*

- Improved API response times by 30% through **object-oriented microservices**, indexing, caching, pagination, and filtered payloads while debugging integrations across Windows, Linux, and macOS production environments reliably at scale.
- Automated builds and 20+ unit/integration suites through **Jenkins, GitHub, PyTest, and Mockito**, generating failure reports and coverage summaries across Linux, Windows, and macOS with repeatable nightly workflows.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) | *India*

- Synchronized 2M+ records through an eight-service **React and Node.js platform** covering authentication, resources, customers, billing, leads, permissions, and notifications with WebSockets and Azure SQL across locations and workflows.
- Reduced resource wastage by 42% through **role-based access and real-time synchronization**, implementing WebSocket updates, automated tests, Azure monitoring, and Docker/Jenkins deployments across eight modular product service domains.

LST Capitals | *Web Developer Intern* | *India* **November 2022 – January 2023**

- Delivered **sub-200ms Flask APIs** across 50K+ booking records, implementing pricing, validation, payment webhooks, Firebase synchronization, and database transactions while preventing conflicting reservations across concurrent customer workflows reliably.
- Reduced manual refund handling by approximately 70% through **payment workflow automation**, supporting refunds, disputes, and consistent status while presenting architecture and backend integration workflows directly to clients reliably.

Skills

Languages: Python, C++, Java, TypeScript, JavaScript, SQL, Shell
Backend & Design: FastAPI, Flask, Django, REST APIs, Object-Oriented Design, Microservices, API Design, Design Patterns
Systems & Data: Linux, Bash, PostgreSQL, MySQL, MongoDB, Redis, AWS S3, AWS RDS
Frontend & Delivery: React, Node.js, HTML, CSS, Docker, Kubernetes, Git, GitHub
Quality & Operations: PyTest, Mockito, Unit Tests, Integration Tests, Test Automation, CI/CD, Jenkins, Debugging, Logging, Observability

Projects

Compliance Case Automation Service - Design Concept

- Designed a **Python compliance workflow** concept with configurable rules, case queues, evidence capture, role-based access, immutable audit history, scheduled checks, exception handling, and documented reviewer decisions for maintainable operations.

Agentic Code Verification Harness - Design Concept

- Designed an **agent-assisted verification harness** concept that scopes coding tasks, checks generated diffs, runs unit and acceptance suites, flags unsafe changes, records evaluations, and requires accountable human approval.

FinNexus | Live

- Built a **secure fintech platform** supporting 50K+ users and 1M+ transactions with Plaid bank linking, token-based access controls, real-time synchronization, advanced filtering, Kubernetes deployment, and MongoDB-backed data access.